BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

LESSON 192 · byVal, byRef and modular programming

Name: _____ Class: _____ Date: _____

Time: _____ Teacher: _____

Work through the pages in order. Answer in BLACK PEN first. Check in GREEN PEN only after the Exit Ticket.

Do Now · Recall previous learning

Q1. What does return do in a C# function?

Q2. Can a method read a local variable declared inside another method?

Q3. What is a[0] for int[] a={9,6,3}?

Q4. If Double(n) returns 2*n, find Double(Double(3)).

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

I Do · Read new knowledge

Key vocabulary

by value — The subroutine receives a copy of the value.

by reference — The parameter refers to the caller's variable.

modularity — Divide a program into smaller testable subroutines.

ref — C# keyword enabling reference parameter passing.

side effect — A change to external program state.

New learning

CALLER — The caller declares marks = 10.

ALIAS — ref makes score refer to the caller's marks rather than an independent copy.

AFTER — Changing score therefore changes marks, which becomes 15.

Worked example

A function receives a copy of x=7 and adds 4. What is x afterwards?

1. Predict first — A new parameter value receives a copy of the integer 7.

2. Model the result — value becomes 11 inside AddFour, but x remains 7 after the call.

We Do · Practise with guidance

A procedure adds 3 to a ref parameter. Trace y=8 through the call.

```
static void AddThree(ref int n)
{
    n += 3;
}
int y = 8;
AddThree(ref y);
```

Predict first

y starts at 8 and is passed using ref.

Model the result

n refers to y, so adding 3 makes y = 11.

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

You Do · Work independently

BLACK PEN: Write your own answers. Do not look at the Review & Improve pages yet.

Question 1

A method takes int number by value and adds 2. Caller x=5. What is x after the call?

Question 2

Write a C# procedure that doubles an integer passed by reference.

Question 3

Explain when returning a result is preferable to changing a ref parameter.

Exit Ticket · Answer before checking

Changing a by-value int parameter always changes the caller's int.

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Review & Improve · GREEN PEN

Now change to GREEN PEN. Find each original question. Tick correct answers; read the model and rewrite mistakes.

Page 1 onwards · Do Now

Q1. Finishes the invocation and returns a result to its caller.

Q2. Not directly. It is outside the variable's scope.

Q3. 9.

Q4. Double(3)=6; Double(6)=12.

You Do pages · Independent questions

Q1. 5; only the parameter copy is updated.

Q2. static void Double(ref int n) { n *= 2; } Call with Double(ref x);

Q3. When only one result is needed, a return makes the data flow explicit and avoids an unexpected side effect on a caller variable.

Exit Ticket · Model answer

False. A by-value parameter has its own copy.

Try the question again

Choose one question you got wrong. All correct? Choose the hardest.

1. Copy the whole original question:

2. Write a complete new answer in GREEN PEN. Show working if needed: